第六章 全同粒子

§ 6.1 Bose / Fermi子

粒子1：$|k\rangle$，粒子2：$|k'\rangle$

两粒子系：$|k\rangle \otimes |k'\rangle$，$|k\rangle|k'\rangle$

对全同粒子：$|k\rangle|k'\rangle$ 与 $|k'\rangle|k\rangle$ 无法分辨

态写为 $|\psi\rangle = c_1|k\rangle|k'\rangle + c_2|k'\rangle|k\rangle$

定义交换算符：$\hat{P}|k\rangle|k'\rangle = |k'\rangle|k\rangle$

$\hat{P}|\psi\rangle = e^{i\theta}|\psi\rangle$ （全同粒子态交换不变）

$e^{i\theta}(c_1|k\rangle|k'\rangle + c_2|k'\rangle|k\rangle) = c_1|k'\rangle|k\rangle + c_2|k\rangle|k'\rangle$

解得 $\theta = 0, \pi$，$c_1 = c_2$ 或 $c_1 = -c_2$

构造全同粒子态：$|\psi_\pm\rangle = \frac{1}{\sqrt{2}}(|k\rangle|k'\rangle \pm |k'\rangle|k\rangle)$，分别对应（交）交换对称

全同粒子 $\begin{cases} \text{Bose子} & \text{整数自旋} \quad \text{交换对称} \\ \text{Fermi子} & \text{半整数自旋} \quad \text{交换反对称} \end{cases}$

对全同粒子系统，$[\hat{P}, H] = 0$

Fermi子：两个全同费米子不能占据同一个单粒子态。

$k = k' \Rightarrow$ Pauli不相容原理 $|\psi\rangle = \frac{1}{\sqrt{2}}(|k\rangle|k'\rangle - |k'\rangle|k\rangle) = 0$

N个Fermi子：

设占据 $k_1 \cdots k_N$ 个单粒子态

$$|\psi\rangle = \frac{1}{\sqrt{N!}}\sum_P (-1)^{n_P}\hat{P}(|k_1\rangle|k_2\rangle\cdots|k_N\rangle)$$
$$= \frac{1}{\sqrt{N!}}\sum_P (-1)^{n_P}\hat{P}|k_1 k_2 \cdots k_N\rangle$$

选取坐标表象下的基矢：

$$|r_1, r_2 \cdots r_N\rangle = |r_1\rangle|r_2\rangle\cdots|r_N\rangle$$

$$|\psi\rangle = \frac{1}{\sqrt{N!}}\int d^3r_1 \cdots d^3r_N \sum_P (-1)^{n_P}\langle r_1, r_2 \cdots r_N|\hat{P}|k_1, k_2, \cdots k_N\rangle|r_1 \cdots r_N\rangle$$
$$= \int d^3r_1\, d^3r_2 \cdots d^3r_N\, \psi_{k_1 k_2 \cdots k_N}(r_1, \cdots r_N)|r_1, r_2 \cdots r_N\rangle$$

其中 $\psi_{k_1 k_2 \cdots k_N}(r_1, \cdots r_N) = \begin{vmatrix} \phi_{k_1}(r_1) & \phi_{k_2}(r_1) & \cdots & \phi_{k_N}(r_1) \\ \phi_{k_1}(r_2) & & & \\ \vdots & & & \\ \phi_{k_1}(r_N) & & & \end{vmatrix}$

Slater 行列式

N 个 Bose 子：

可以有任意多个 Bose 子占据同一个单粒子态

N 个 Bose 子占据 m 个单粒子态，设各态上粒子数为 $n_1 \cdots n_m$

$N = n_1 + n_2 + \cdots + n_m$

$$|\psi\rangle = \sqrt{\frac{\prod n_i!}{N!}} \sum_P \hat{P} |1 \cdots 1 m_2 \cdots\rangle$$

§6.2 两电子系统

两电子系统态：

$$|\Psi\rangle = \sum_{s_{12}, s_{22}} \int d^3r_1\, d^3r_2\, \Psi(r_1, s_{12}, r_2, s_{22}) |r_1, s_{12}\rangle |r_2, s_{22}\rangle$$

$$|\Psi\rangle = \sum_{s_{12}, s_{22}} \int d^3r_1\, d^3r_2\, \psi(r_1, r_2)\, \chi(s_{12}, s_{22}) |r_1, r_2\rangle |s_{12}, s_{22}\rangle$$

$$= \sum_{s, s_2} \int d^3r_1\, d^3r_2\, \psi(r_1, r_2) |r_1, r_2\rangle |s, s_2\rangle \qquad \vec{S} = \vec{S_1} + \vec{S_2},\ \vec{S_2} = \vec{s_{12}} + \vec{s_{22}}$$

假定两电子占据两单粒子态 $|k_1\rangle, |k_2\rangle$ 电子为费米子，波函数反对称.

(1) 自旋单重态 $|\Psi\rangle = \frac{1}{\sqrt{2}} (|k_1\rangle|k_2\rangle + |k_2\rangle|k_1\rangle) |00\rangle$

(2) 自旋三重态 $|\Psi\rangle = \frac{1}{\sqrt{2}} (|k_1\rangle|k_2\rangle - |k_2\rangle|k_1\rangle)$ $\begin{cases} |11\rangle \\ |10\rangle \\ |1-1\rangle \end{cases}$

例：考虑一维势井 $V(x) = \begin{cases} 0 & 0 \le x \le a \\ \infty & x<0,\ x>a \end{cases}$ 势井中有两个全同粒子

(1) 如果粒子自旋 $s=0$，写出体系最低两个能级，指出简并度，给出波函数，

$s=0$ 为 Boson 子，且自旋部份又有交换对称一种情况

基态：两个粒子都处在最低能级.

$$\varepsilon_0 = 2E_1 = \frac{\pi^2\hbar^2}{ma^2} \qquad \psi_1(x) \text{ 为第一能级的空间波函数}$$

$$\psi_0(x_1, x_2) = \psi_1(x_1)\, \psi_1(x_2) |00\rangle$$

第一激发态：一个在最低，一个在次低

$$\varepsilon_0 = E_1 + E_2 = \frac{5\pi^2\hbar^2}{2ma^2}$$

$$\psi_1(x_1, x_2) = \frac{1}{\sqrt{2}} \left[\psi_1(x_1)\psi_2(x_2) + \psi_2(x_1)\psi_1(x_2)\right] |00\rangle$$

(2) 如果自旋 $s=\frac{1}{2}$ 呢？ $s=\frac{1}{2}$ 为 Fermi 子

基态：$\varepsilon_0 = 2E_1 = \frac{2\pi^2\hbar^2}{ma^2}$ 空间部份对称，自旋部份反对称

$$\Psi_0(x_1, x_2) = \psi_1(x_1)\, \psi_1(x_2) |00\rangle$$

第一激发态：$\varepsilon_1 = E_1 + E_2 = \frac{5\pi^2\hbar^2}{2ma^2}$

$$\Psi_1(x_1, x_2) = \begin{cases} \frac{1}{\sqrt{2}} \left[\psi_1(x_1)\psi_2(x_2) + \psi_2(x_1)\psi_1(x_2)\right] |00\rangle \\ \frac{1}{\sqrt{2}} \left[\psi_1(x_1)\psi_2(x_2) - \psi_2(x_1)\psi_1(x_2)\right] \begin{cases} |11\rangle \\ |10\rangle \\ |1-1\rangle \end{cases} \end{cases}$$

§6.3 交换相互作用

考虑两个电子占据两个正交轨道 $\phi_\alpha(r)$, $\phi_\beta(r)$, 对应能量 $\varepsilon_\alpha, \varepsilon_\beta$

$$\hat{H} = \hat{H}_0(r_1) + \hat{H}_0(r_2) + \hat{H}_{int}(|r_1 - r_2|)$$

$$= \frac{\hat{p}_1^2}{2m} + V(r_1) + \frac{\hat{p}_2^2}{2m} + V(r_2) + \frac{1}{4\pi\varepsilon_0}\frac{e^2}{|r_1 - r_2|}$$

自旋单重态: $\psi_s(r_1, r_2) = \frac{1}{\sqrt{2}}\left\{\phi_\alpha(r_1)\phi_\beta(r_2) - \phi_\beta(r_1)\phi_\alpha(r_2)\right\}$ 空间部分反对称

自旋三重态: $\psi_a(r_1, r_2) = \frac{1}{\sqrt{2}}\left\{\phi_\alpha(r_1)\phi_\beta(r_2) + \phi_\beta(r_1)\phi_\alpha(r_2)\right\}$ 空间部分对称

$\langle \psi_s | \hat{H} | \psi_s \rangle = \varepsilon_\alpha + \varepsilon_\beta + U_{\alpha\beta} + J_{\alpha\beta}$

$\langle \psi_a | \hat{H} | \psi_a \rangle = \varepsilon_\alpha + \varepsilon_\beta + U_{\alpha\beta} - J_{\alpha\beta}$

其中 $U_{\alpha\beta} = \frac{e^2}{4\pi\varepsilon_0}\int d^3r_1\, d^3r_2 \frac{|\phi_\alpha(r_1)|^2 |\phi_\beta(r_2)|^2}{|r_1 - r_2|}$

$J_{\alpha\beta} = \frac{e^2}{4\pi\varepsilon_0}\int d^3r_1\, d^3r_2 \frac{\phi_\alpha^*(r_1)\phi_\beta(r_1)\phi_\alpha(r_2)\phi_\beta^*(r_2)}{|r_1 - r_2|}$

推导: $\langle \psi_s | \hat{H} | \psi_s \rangle$

$= \int d^3r_1\, d^3r_2\, \psi_s^*(r_1, r_2)\, \hat{H}\, \psi_s(r_1, r_2)$

$= \int d^3r_1\, d^3r_2\, \frac{1}{\sqrt{2}}\left\{\phi_\alpha^*(r_1)\phi_\beta^*(r_2) + \phi_\beta^*(r_1)\phi_\alpha^*(r_2)\right\}\left\{\frac{\hat{p}_1^2}{2m} + V(r_1) + \frac{\hat{p}_2^2}{2m} + V(r_2) + \frac{1}{4\pi\varepsilon_0}\frac{e^2}{|r_1 - r_2|}\right\}$

$\frac{1}{\sqrt{2}}\left\{\phi_\alpha(r_1)\phi_\beta(r_2) + \phi_\beta(r_1)\phi_\alpha(r_2)\right\}$

$= \frac{1}{2}\int d^3r_1\, d^3r_2 \left\{\phi_\alpha^*(r_1)\left\{\frac{\hat{p}_1^2}{2m} + V(r_1)\right\}\phi_\alpha(r_1)\phi_\beta^*(r_2)\phi_\beta(r_2) = \varepsilon_\alpha\right.$

$+ \phi_\beta^*(r_1)\phi_\beta(r_1)\phi_\alpha^*(r_2)\left\{\frac{\hat{p}_2^2}{2m} + V(r_2)\right\}\phi_\alpha(r_2) = \varepsilon_\alpha$

$+ \phi_\alpha^*(r_1)\phi_\alpha(r_1)\phi_\beta^*(r_2)\left\{\frac{\hat{p}_2^2}{2m} + V(r_2)\right\}\phi_\beta(r_2)$

$+ \phi_\beta^*(r_1)\left\{\frac{\hat{p}_1^2}{2m} + V(r_1)\right\}\phi_\beta(r_1)\phi_\alpha^*(r_2)\phi_\alpha(r_2)\Big\}$

$+ \phi_\alpha^*(r_1)\left\{\frac{\hat{p}_1^2}{2m} + V(r_1)\right\}\phi_\beta(r_1)\phi_\beta^*(r_2)\left\{\frac{\hat{p}_2^2}{2m} + V(r_2)\right\}\phi_\alpha(r_2)$ 交叉项 = 0

$+ \phi_\beta^*(r_1)\left\{\frac{\hat{p}_1^2}{2m} + V(r_1)\right\}\phi_\alpha(r_1)\phi_\alpha^*(r_2)\left\{\frac{\hat{p}_2^2}{2m} + V(r_2)\right\}\phi_\beta(r_2)$ = 0

$+ \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0}\int d^3r_1\, d^3r_2 \left\{\phi_\alpha^*(r_1)\phi_\alpha(r_1)\phi_\beta^*(r_2)\phi_\beta(r_2) + \phi_\beta^*(r_1)\phi_\beta(r_1)\phi_\alpha^*(r_2)\phi_\alpha(r_2)\right.$ $U_{\alpha\beta}$

$\left.+ \phi_\alpha^*(r_1)\phi_\beta(r_1)\phi_\beta^*(r_2)\phi_\alpha(r_2) + \phi_\beta^*(r_1)\phi_\alpha(r_1)\phi_\alpha^*(r_2)\phi_\beta(r_2)\right\}\frac{1}{|r_1 - r_2|}$ $J_{\alpha\beta}$

$\frac{e^2}{4\pi\varepsilon_0|r_1-r_2|}$ α β α β

$J_{\alpha\beta}$ $+\psi_\beta^*(r_1)\psi_\alpha(r_1)\psi_\alpha^*(r_2)\psi_\beta(r_2)+\psi_\beta^*(r_1)\psi_\beta(r_1)\psi_\alpha^*(r_2)\psi_\alpha(r_2)\Big|$ $u_{\alpha\beta}$

两交换耦合作用空间有效交换作用的本征值.

$(\vec{S}_1+\vec{S}_2)^2=\vec{S}_1^2+\vec{S}_2^2+2\vec{S}_1\cdot\vec{S}_2=\frac{3}{2}\hbar^2+2\vec{S}_1\cdot\vec{S}_2$

有效交换作用: $\vec{S}_1\cdot\vec{S}_2$

$\langle\psi_s|\hat{H}|\psi_s\rangle=\varepsilon_\alpha+\varepsilon_\beta+u_{\alpha\beta}-\frac{1}{2}J_{\alpha\beta}-2J_{\alpha\beta}\langle\psi_s|\frac{1}{\hbar^2}\vec{S}_1\cdot\vec{S}_2|\psi_s\rangle$

$\langle\psi_a|\hat{H}|\psi_a\rangle=\varepsilon_\alpha+\varepsilon_\beta+u_{\alpha\beta}-\frac{1}{2}J_{\alpha\beta}-2J_{\alpha\beta}\langle\psi_a|\frac{1}{\hbar^2}\vec{S}_1\cdot\vec{S}_2|\psi_a\rangle$

推导: $\langle\psi_s|\vec{S}_1\cdot\vec{S}_2|\psi_s\rangle$

$=\frac{1}{2}\langle\psi_s|\vec{S}^2-\vec{S}_1^2-\vec{S}_2^2|\psi_s\rangle$

$=\frac{1}{2}\left(2-\frac{3}{4}-\frac{3}{4}\right)\hbar^2=-\frac{3}{4}\hbar^2$

$\langle\psi_a|\vec{S}_1\cdot\vec{S}_2|\psi_a\rangle=\frac{1}{2}\left(2-\frac{3}{4}-\frac{3}{4}\right)\hbar^2=\frac{1}{4}\hbar^2$

↳ $\hat{H}_{ex}=-2J\left(\frac{1}{\hbar^2}\vec{S}_1\cdot\vec{S}_2+\frac{1}{4}\right)$

$J>0$ 铁磁交换作用

$J<0$ 反铁磁交换作用

对原子轨道上的两电子

$J_{\alpha\beta}=\frac{e^2}{4\pi\varepsilon_0}\int d^3r_1\,d^3r_2\,\frac{\Phi^*(r_1)\Phi(r_2)}{|r_1-r_2|}$ $\Phi(r_2)=\psi_\alpha(r_2)\psi_\beta^*(r_2)$

$=\frac{e^2}{4\pi\varepsilon_0}\int\frac{d^3k}{(2\pi)^{3/2}}d^3r_1\,\Phi^*(r_1)\Phi(k)\int d^3r_2\,\frac{e^{i\vec{k}\cdot\vec{r}_2}}{|\vec{r}_1-\vec{r}_2|}$ $\Phi(r_2)=\frac{1}{(2\pi)^{3/2}}\int d^3k\,\Phi(k)e^{i\vec{k}\cdot\vec{r}_2}$

$=\frac{e^2}{4\pi\varepsilon_0}\int\frac{d^3k}{(2\pi)^{3/2}}\frac{4\pi}{k^2}\Phi(k)\int d^3r_1\,\Phi^*(r_1)e^{i\vec{k}\cdot\vec{r}_1}$ $\int\frac{e^{i\vec{k}\cdot\vec{r}}}{|\vec{r}|}=\frac{4\pi}{k^2}$

$=\frac{e^2}{4\pi\varepsilon_0}\int\frac{d^3k}{(2\pi)^{3/2}}\frac{4\pi}{k^2}|\Phi(k)|^2>0$ $\Phi^*(k)=\frac{1}{(2\pi)^{3/2}}\int d^3r\,\Phi^*(r)e^{i\vec{k}\cdot\vec{r}}$

⇒ 原子内不同轨道电子自旋之间铁磁交换作用.